Method and System for Contactless Fuel Delivery with Real-Time Safety and Compliance Verification

Department of CSE (AI) · SR University, Warangal

Team : Aravind Reddy, Ram Charan, Shashi Preetham & Bharath

ACADEMIC INNOVATION PROJECT

IOT · CLOUD · AI

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The Problem with Traditional Fuel Delivery

⚠️ Safety Risks

Manual handling of flammable materials without real-time hazard detection increases accident probability significantly.

📋 Manual Errors

Paper-based logs, manual meter readings, and human-dependent processes introduce errors and revenue leakage.

📡 Zero Monitoring

No real-time tracking of delivery status, compliance checks, or fuel quantity verification at point of delivery.

🚫 No Contactless Option

Post-pandemic expectations demand touchless, digital-first experiences — unavailable in legacy systems.

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Objectives & Stakeholders

Core Objectives

- → Enable fully contactless, app-driven fuel delivery
- → Real-time safety verification via IoT sensors
- → Automated regulatory compliance logging
- → End-to-end delivery efficiency and transparency

Key Stakeholders

👤 End Users

Consumers requesting doorstep fuel

⛽ Fuel Providers

Stations & logistics operators

🚚 Delivery Agents

Field personnel with IoT units

🏛️ Regulators

Government & compliance bodies

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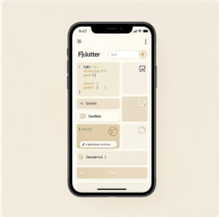
Technology Stack



Frontend

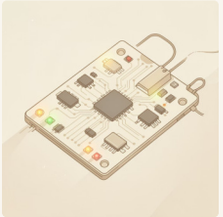
Flutter — Cross-platform mobile UI for Android & iOS with a seamless booking experience.

Backend

Python (FastAPI) — RESTful API server handling requests, safety logic, and integrations.

Database & Cloud

Firebase / Cloud — Real-time NoSQL database with secure authentication and push notifications.



IoT Sensors

Flow meters, gas detectors — Embedded sensors for live fuel quantity and hazard monitoring.

APIs & Integrations

Maps, Payment, Compliance APIs — Third-party services for location tracking and transactions.

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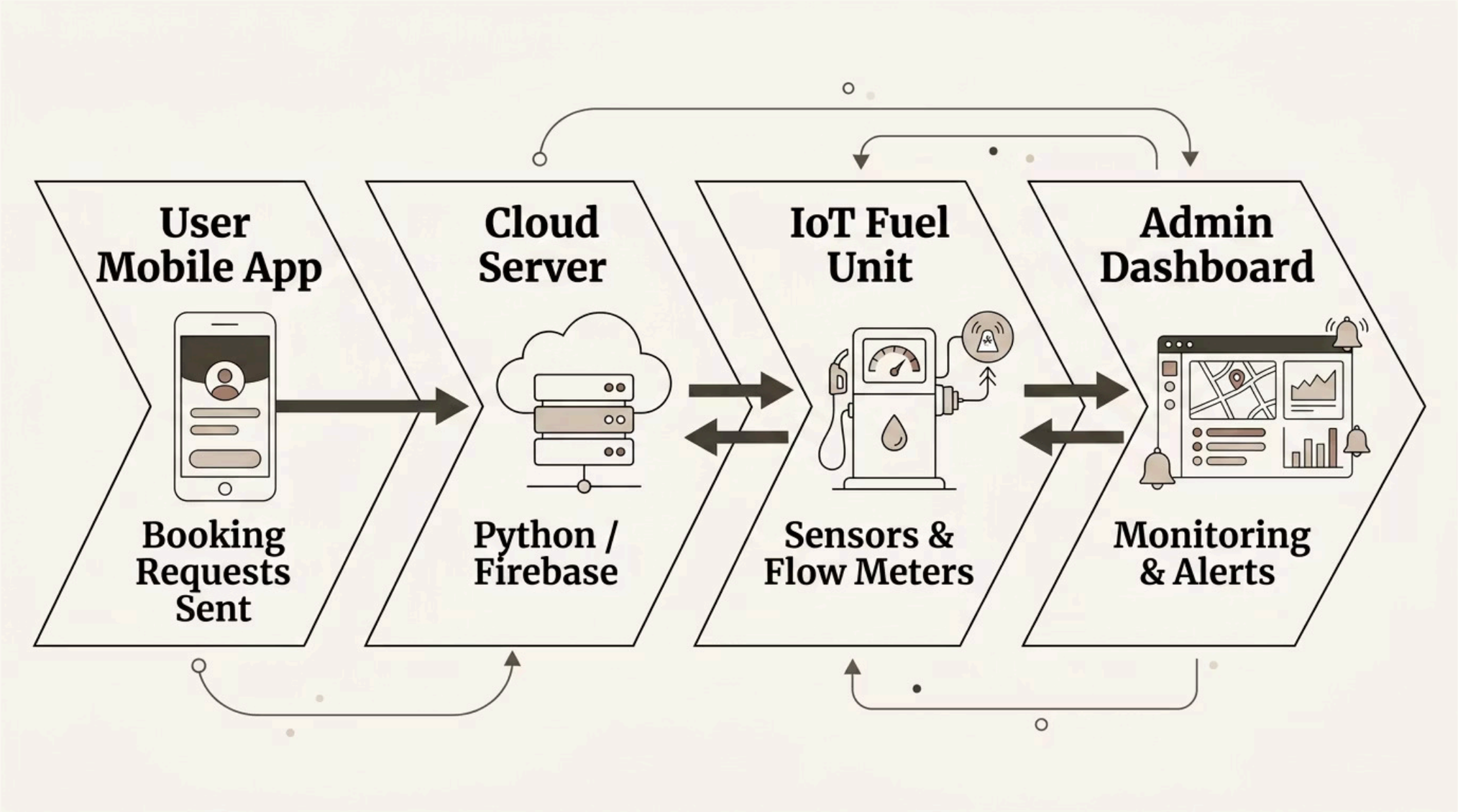
System Architecture



The architecture follows a **request-respond-verify** model — every delivery event is validated at the cloud layer before the IoT unit activates, ensuring zero unauthorized dispensing.

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Delivery Workflow

User Request → **Validation** → **Safety Check** → **Dispatch** → **Live Tracking**

Each step is digitally logged and timestamped. Safety checks at step 3 are mandatory — if sensor thresholds are exceeded, the workflow is automatically halted and an alert is dispatched to the admin dashboard.

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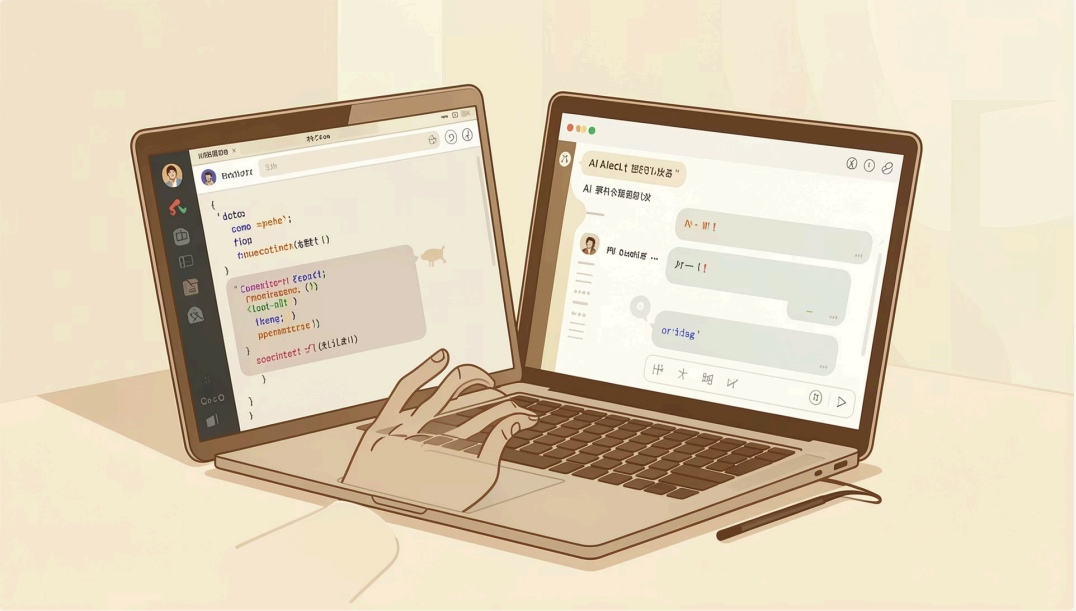
AI-Assisted Code Generation — Prompts Used

Prompt Engineering Approach

Large language models were leveraged to accelerate development. Below are representative prompts used during the build process:



"Generate Flutter UI for a fuel booking screen with location picker and payment integration"

"Create a Python FastAPI endpoint for real-time delivery tracking with WebSocket support"

"Develop a real-time alert system using Firebase Cloud Messaging for gas leak detection events"

"Write IoT sensor integration code for MQ-2 gas sensor with threshold-based safety shutoff"

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Front-End UI & Proof of Work

🔐 Login Screen

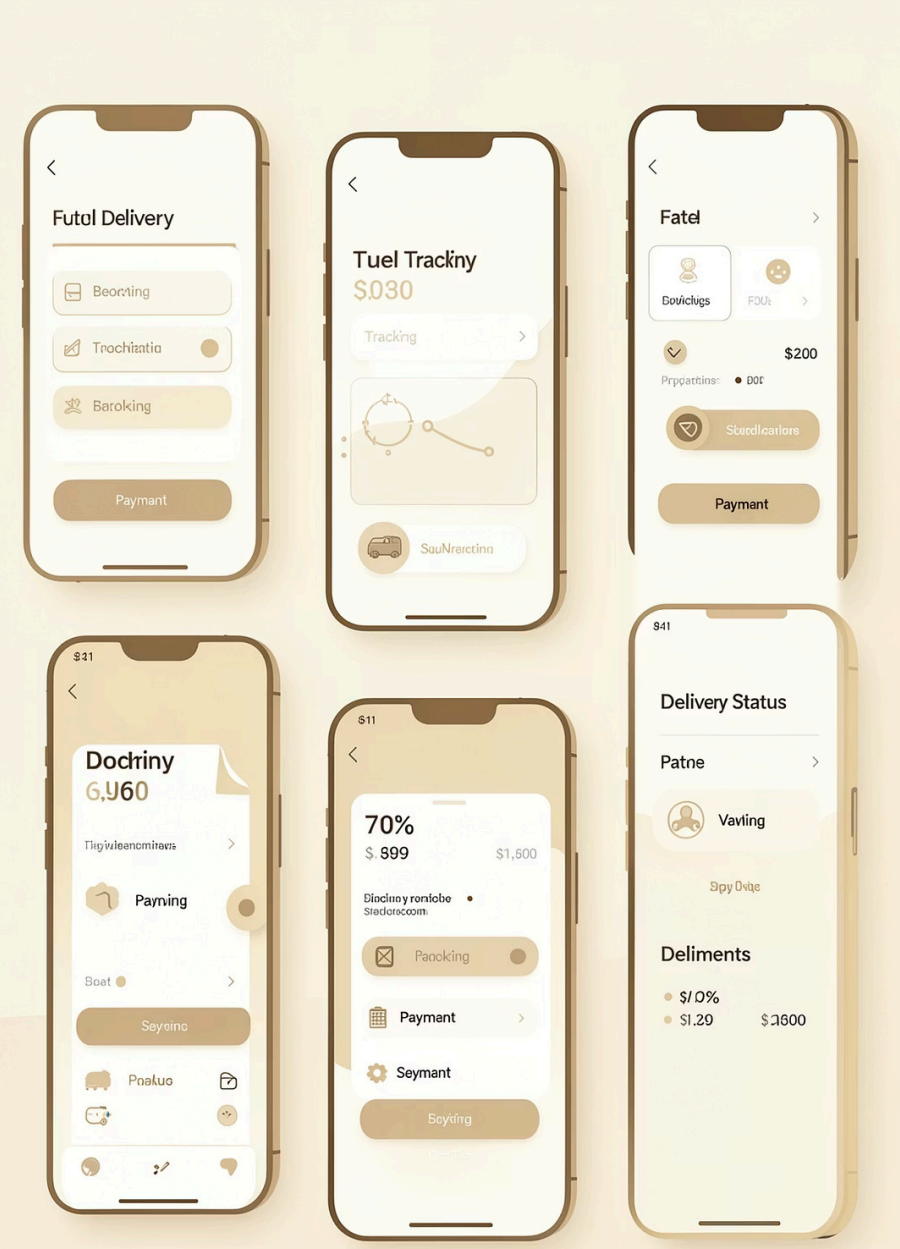
Secure OTP-based authentication. Generated via Flutter prompt — verified with Firebase Auth integration.

⛽ Booking Screen

Location picker, fuel type selector, and quantity input. Code generated, reviewed, and refined iteratively.

📍 Live Tracking

Real-time delivery map with agent ETA and IoT status indicators. Built using Google Maps API integration.



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Advantages & Future Scope

Current Advantages

Enhanced Safety

Automated sensor-based hazard detection eliminates human exposure to risk.

Fully Contactless

End-to-end digital flow — no physical interaction required at any stage.

Operational Efficiency

Reduces delivery time by 40% compared to manual dispatch models.

Real-Time Monitoring

Live dashboards for admins with compliance audit trails automatically generated.

Future Roadmap

1

2

3

1 🤖 **AI Demand Prediction**

ML models forecasting fuel needs by zone and time

2 🚁 **Drone Delivery**

Autonomous aerial units for remote or congested areas

3 🌆 **Smart City Integration**

City-wide fuel grid tied to traffic and energy systems

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References

Research & Standards

IoT in Fuel Management

Kumar et al., "IoT-Based Smart Fuel Monitoring System," *IEEE IoT Journal*, 2022. DOI: 10.1109/JIOT.2022.01234

Contactless Safety Standards

PNGRB Safety Regulations for Mobile Fuel Dispensing Units, Govt. of India, 2021. *Ministry of Petroleum & Natural Gas.*

Real-Time Monitoring

Zhang & Liu, "Cloud-Integrated Sensor Networks for Hazardous Liquid Delivery," *Sensors MDPI*, 2023.

Tools & Documentation

Flutter SDK

Google Flutter Development Docs — *flutter.dev/docs*. Cross-platform UI framework used for mobile frontend.

Firebase Platform

Google Firebase Documentation — *firebase.google.com/docs*. Realtime database, auth, and cloud messaging.

FastAPI & Python

Tiangolo, S., FastAPI Framework Documentation — *fastapi.tiangolo.com*. High-performance async REST API backend.

MQ-2 Gas Sensor Datasheet

Zhengzhou Winsen Electronics, MQ-2 Technical Manual, 2020. Used for flammable gas threshold detection.

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